

Principal plane of APP

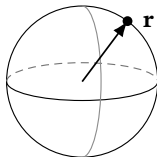
Pauli representation

Bloch sphere

$$\begin{aligned} S^\top &= S, & J^\top &= -J \\ S^2 &= I, & J^2 &= -I \\ SJ &= -JS \end{aligned}$$



$$\begin{aligned} \sigma_1 &= SJ \\ \sigma_2 &= -iJ \\ \sigma_3 &= S \end{aligned}$$



complexification

pure states

$$\rho_\psi = \psi\psi^\dagger = \frac{1}{2}(I + \mathbf{r} \cdot \boldsymbol{\sigma})$$

$$h : r_k = \psi^\dagger \sigma_k \psi$$

$$S^1 \quad \xrightarrow{\quad \quad \quad}$$

$$S^3 \quad \xrightarrow{\quad \quad \quad}$$

$$S^2 \simeq \mathbb{CP}^1$$

common phase
 $e^{i\chi}\psi$

unit vector
 $\psi^\dagger\psi = 1$

ray
 $\|\mathbf{r}\| = 1$